

Notes

Three Ideas Each

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- Audio Spectrum Motor Health Check - Use a directional mic (or contact mic) placed near each motor while spooling. Perform FFT to detect abnormal harmonics/roughness signatures vs a baseline. Output: PASS/FAIL + "bearing/imbalance suspected."
- <https://rotorvillage.ca/geprc-speedx2-1404-4600kv-motor/>
- The idea is to use the intuitive approach that the drone handler already does where he listens to the motor and make it a more objective

Hasnain

1. Software analysis of the open radio frequencies that would be safe to use in an urban environment after considering radio frequencies are emitted by high power wifi routers or smart devices
 - a. Measuring all the radio frequencies across a large perimeter is called RF spectrum survey [1] (Probably not possible due to budget restraints. Sensors needed to scan such a wide area are in the thousands of dollars, and using multiple low cost sensors and doing a manual sweep would take too long. Would never meet the time requirement which is an absolute restriction and cannot be adjusted whatsoever) (Look into alternatives to RF spectrum survey for maybe cheaper opinion)
 - i. Uses spectrum analyzer or Software Defined Radio
 - ii. Gives a f_{start} and f_{stop} for a wideband picture of all active signals at any given location
 - iii. Two types of sensors:
 1. Swept-tuned sensors: (This is not ideal. This is lab grade and is overkill here. These are also very localized and not needed for large area scanning)
 - a. traditional superheterodyne swept-tuned spectrum analyzer. The swept-tuned spectrum analyzer downconverts an RF signal to a lower intermediate frequency (IF), typically using several frequency conversion stages
 2. DFT-Based Sensor: (the preferred type due to its ability to scan a large range)
 - a. A type of Software Defined Radio (SDR) that digitizes incoming RF signals and applies a Discrete Fourier Transform (DFT) to produce an amplitude vs. frequency output

- b. All local oscillators are set to fixed frequencies; to cover a wide frequency range, multiple DFT captures are taken and "stitched together"
- c. Their behavior can be fully changed via software — no new hardware needed to change operating frequencies or bandwidth
- d. Possible sensor we could use at a reasonable price point is: TinySa Ultra
 - i. However even this will probably struggle in very dense urban environments because of all the noise
 - ii. Will probably need a real time spectrum analyzer to have any sort of effective scan but that is way out of budget.

RF analysis:

1. <https://its.ntia.gov/publications/download/tr-15-519.pdf>
2. <https://arxiv.org/pdf/2301.02380>
3. https://www.ieee.li/pdf/viewgraphs/rf_sensor_networks_%20for_improved_detection_geo_location.pdf
4. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10490811/>
 - a. Drone Detection and Tracking Using RF Identification Signals
 - i. Highly relevant???
5. <https://www.rtl-sdr.com/tinysa-ultra-reviews-a-0-1-mhz-6-ghz-spectrum-analyzer-for-120/>
6. <https://www.aliexpress.com/item/4001274404758.html>
7. https://www.reddit.com/r/sdr/comments/1iwpvx0/tinysa_or_hackrf_for_spectrum_analysis_upto_few/
 - a. Context to help understand but not reliable source. Info needs to be verified
8. <https://signalhound.com/products/page/2/>
 - a. Real time spectrum analyzer (wayyy out of budget)



SA44B — 4.4 GHz Spectrum Analyzer



\$1,295.00 – \$1,745.00

2. How to analyze defects in Carbon Fiber (Look at Formula 1?)
 - a. Ultrasonic frequency measurement of the material
 - i. Unreliable without the proper equipment. Using DIY equipment would give questionable results at best. The reading instruments would not have enough accuracy to find microfractures and would find defects when it is too late
 - b. Natural frequency of material comparison
 - i. HIGHLY dependent on setup conditions, and the frame. Any minor changes like if it is not held in the exact same way, the battery is not in the exact same position, etc. the test would not be reliable.
3. How is battery health analyzed? Look at smart phone batteries. How is their health analyzed and how accurate is it?
 - a.
4. Motor analysis based on audio. Use higher accuracy audio sensors. How else can motors have defects
 - a. Voltage Analysis
 - b. Sound Analysis
 - i. Using a mini sound chamber to cut down on external noise
 - c. Torque Analysis
 - i.
5. Regarding goal 6, perhaps setup the diagnostic system with an ability to wait for wifi connection and then update using the local log created at time of testing to be uploaded into a cloud log
 - a. This must be done regardless

Michael

Torque Symmetry Score

- DC motor can be used to check how evenly torque is produced during rotation
- Torque production needs to be consistent for healthy motors
- Uneven torque could represent debris or misalignment of the motor
-

It's a **numerical metric** that tells you whether a motor is behaving evenly and smoothly during rotation.

In a healthy BLDC motor:

- Each phase produces equal torque
- Rotation is smooth
- Acceleration curve is consistent
- No periodic "catching" or resistance

If something is wrong (bearing wear, debris, bent shaft, partial winding damage), torque output becomes uneven.

That unevenness shows up as:

- Oscillations in angular acceleration
- Current ripple irregularity
- Periodic speed fluctuation

So a "Torque Symmetry Score" could be something like:

$$\text{Symmetry Score} = \frac{\text{Standard deviation of torque ripple}}{\text{Mean torque}}$$

Low score → smooth motor

High score → mechanical imbalance

You're basically quantifying:

“Does this motor spin like a perfect circle, or like a slightly dented wheel?”

Can be done on a bench

Method A — Current Ripple Analysis (Simplest)

Run motor at fixed RPM.

Measure:

- Input current waveform
- Current ripple amplitude
- Periodic variation

Uneven torque → uneven current draw.

No extra sensors required.

Just a current shunt + ADC.

Method B — Angular Acceleration Consistency (Stronger)

Add a small magnetic encoder to measure RPM precisely.

If torque is uneven, you'll see:

- Micro-speed fluctuations
- Tiny periodic slow-downs
- Irregular acceleration slope

This directly measures mechanical symmetry.

Method C — Coast-Down Decay Test (Very Elegant)

Spin motor to fixed RPM.

Cut power.

Measure time to stop.

Healthy motor:

- Smooth exponential decay

Damaged motor:

- Faster decay
- Irregular deceleration

You can compute:

$$\tau = \text{time constant} \downarrow \text{rotational decay}$$

That gives a "Drain Index"

Motor Analysis (all AI)

Frank

Fault types in BLDC motors (secondary research):

FPV drone motors aGEPRC Cinelog25 V2 HD O3 Quadcopter Brushless DC (BLDC) motors. Academic literature identifies several well-defined fault categories relevant to field diagnostics:

1. Winding faults (inter-turn shorts, open circuits) A multimeter winding resistance test is the standard first check. Measuring phase-to-phase resistance ($A \rightarrow B$, $B \rightarrow C$, $C \rightarrow A$) should yield nearly equal values. According to GetFPV's electrical diagnostics guide, *"most motors should measure around 50 milliohms. Anything lower and you have yourself an excellent quality motor. Anything higher and you have a burnt out or low-quality motor."* (GetFPV Learn, "FPV Drone Electrical Checks Using a Multimeter"). The IntoFPV community forum notes a typical healthy range of *"50–200 mΩ"* between phases, with a reading near 0 indicating a short circuit and infinite resistance indicating an open winding. Brushless.com confirms: *"if the reading shows 0, this indicates a short circuit between the phases. If the reading shows 2000 ohms or infinity, this indicates an open winding."*

Critical limitation to flag: Standard consumer multimeters cannot accurately resolve milliohm-range resistance. GetFPV explicitly warns: *"Unless your multimeter has a lower resistance setting than 200 ohms, it will most likely not be able to measure the motor resistance accurately."* This supports the RFP's framing that current tooling is inadequate for the community — a dedicated precision resistance circuit (or four-wire Kelvin measurement) would be needed.

2. Bearing faults A peer-reviewed study published in *Designs* (Kudelina et al., 2020, "Bearing Fault Analysis of BLDC Motor for Electric Scooter Application") found that bearing faults are detectable through vibration spectrum analysis using FFT. Their experiment showed the fundamental frequency amplitude of faulty bearings was *significantly higher* than healthy ones, with prominent fault signatures appearing in the 15–35 Hz range. This directly validates the approach of using MEMS vibration sensing, which the RFP already proposes via the Analog Devices EVAL-CN0532-EBZ.

3. Vibration analysis and MCSA A paper in *IEEE Access* (Shifat & Hur, 2020, "An effective stator fault diagnosis framework of BLDC motor based on vibration and current signals") confirms that combining vibration signals with current analysis is an effective stator fault diagnosis approach for BLDC motors. Motor Current Signature Analysis (MCSA) — analyzing the current waveform's frequency components — can detect broken rotor bars, winding shorts, and ESC-related anomalies (Heli Tech Energy, "Fault Diagnosis Methods for a Micro DC Brushless Motor"). This is relevant because Danny notes that *"the ESC or the flight controller is pushing too much power to one side"* [2.5] is one cause of the creaking/rattling sound, meaning motor and ESC faults can produce overlapping symptoms.

4. Back-EMF test (continuity of windings) JKONG Motor's testing guide notes that spinning the motor shaft by hand and measuring AC voltage across two leads (back-EMF) confirms

winding integrity without requiring the motor to be powered. This is a passive, non-destructive test performable in a field setting with minimal equipment.

GetFPV Learn — "FPV Drone Electrical Checks Using a Multimeter and Smoke Stopper" <https://www.getfpv.com/learn/fpv-essentials/fpv-drone-electrical-checks/> (Phase resistance benchmark of ~50 mΩ; multimeter accuracy warning)

IntoFPV Community Forum — "Short on Motor Test" <https://intofpv.com/t-short-on-motor-test> (Typical healthy phase resistance range of 50–200 mΩ)

Brushless.com — "How to Test Brushless DC Motor With Multimeter" <https://www.brushless.com/how-to-test-brushless-dc-motor-with-multimeter> (Phase-to-phase resistance thresholds; short circuit and open winding indicators)

Kudelina et al. (2020) — "Bearing Fault Analysis of BLDC Motor for Electric Scooter Application," *Designs*, Vol. 4, No. 4, MDPI <https://www.mdpi.com/2411-9662/4/4/42> (Peer-reviewed; FFT vibration analysis for bearing fault detection in BLDC motors; 15–35 Hz fault frequency range)

Shifat & Hur (2020) — "An Effective Stator Fault Diagnosis Framework of BLDC Motor Based on Vibration and Current Signals," *IEEE Access*, Vol. 8 (Peer-reviewed; combined vibration + current signal approach for BLDC stator fault diagnosis — cited via secondary reference in *ECTI Transactions*)

Heli Tech Energy — "Fault Diagnosis Methods for a Micro DC Brushless Motor" <https://www.helintelligent.com/blog/what-are-the-fault-diagnosis-methods-for-a-micro-dc-brushless-motor-658594.html> (Overview of MCSA, vibration analysis, thermal monitoring, and winding resistance testing)

JKONG Motor — "How to Test a 3 Wire Brushless Motor" <https://www.jkongmotor.com/how-to-test-a-3-wire-brushless-motor.html> (Back-EMF test method; insulation continuity to motor casing)

Tyto Robotics / Unmanned Systems Technology — "How to Test Motor Winding Resistance" <https://www.unmannedsystemstechnology.com/feature/how-to-test-motor-winding-resistance/> (Winding resistance methodology; Ohm's law application)

Internal Resistance (IR) as State of Health (SoH) metric:

Internal Resistance is the primary measurable indicator of a LiPo cell's health. Grepow's drone battery guide states: "A healthy LiPo cell typically has an IR below 10 mΩ; significantly higher values indicate an aging battery that needs replacement." The RFP cites Typhoon Energy's guide establishing the healthy range as 10–15 mΩ per cell [7]. HereWin's maintenance guide adds a concrete threshold: "New batteries have an internal resistance of 15-20 mΩ—be alert if it exceeds 30 mΩ (sign of aging)."

Note on range variability: IR is highly capacity-dependent. RCHelicopterFun's reference guide notes that large, high-capacity, high-C-rating cells can read as low as *"1 to 4 mΩ per cell when brand new"*, while smaller FPV cells will naturally read higher. The absolute threshold matters less than tracking **change over time** from a known baseline — Grepow calls this "The Golden Rule."

Measurement method: The most accurate field method is a four-wire (Kelvin) connection ESR/IR meter, which eliminates lead resistance from the reading. Smart chargers (e.g., iSDT, HOTA, ToolkitRC) offer built-in IR measurement as a more accessible alternative. Most chargers measure AC-IR, which Grepow confirms is *"an excellent and consistent proxy for a battery's health"* despite not being identical to DC-IR.

Temperature as a critical confound: IR is strongly temperature-dependent, making temperature control critical for consistent readings. Grepow states it should always be measured at *"room temperature (~25°C or 77°F) for a reliable baseline."* This intersects directly with the RFP's -15°C field condition requirement — a battery measured warm will show misleadingly low IR.

Cold-temperature performance (electrochemical basis):

The mechanism behind Danny's observation [2.13] is well-documented. HobbyKing's low-temperature LiPo analysis explains: *"In a low temperature environment (usually below 0°C), the viscosity of the electrolyte increases and the activity of lithium ions decreases, resulting in increased internal resistance, reduced output power, and rapid voltage drop."* HereWin provides a concrete data point for ternary lithium cells: IR *"is about 25 mΩ at room temperature, which can soar to over 80 mΩ at -20°C"* — more than a 3× increase. Their report also notes LiPo capacity drops approximately 30% at -20°C, directly matching Danny's reported ~60% usable flight time in the cold.

The FPV Freedom Coalition's beginner guide frames the practical consequence: *"Their voltage will be lower and the sag you feel when pushing them to their limits will be greater. The end result is less power and shorter flight times."*

Voltage Sag:

Voltage sag is the in-flight voltage drop under high-current load (e.g., full throttle). The RFP's Objective 2.2 specifies that healthy cells should maintain >90% of nominal voltage under a simulated discharge load [8]. Grepow's troubleshooting guide identifies the two main triggers as high power demand and cold weather — both directly relevant to the FPV cinematography context. Over-discharge (below 3.0–3.2V per cell) causes permanent chemical damage according to multiple hobbyist and industry sources (FPV Freedom Coalition, Grepow,

vibms.com), reinforcing why pre-flight voltage-sag testing is safety-critical rather than merely informational.

Grepow — "What is the Internal Resistance of a Drone Battery"

<https://www.grepow.com/blog/what-is-the-internal-resistance-of-a-drone-battery.html> (*IR measurement methods; temperature dependence; AC-IR as SoH proxy; "Golden Rule" of tracking change over time*)

Grepow — "Comprehensive Drone Battery Troubleshooting Guide"

<https://www.grepow.com/blog/comprehensive-drone-battery-troubleshooting-guide.html> (*Voltage sag causes; <10 mΩ healthy cell threshold; cold weather IR increase*)

HereWin — "Low-Temp Drone Battery Hacks"

<https://www.herewinpower.com/blog/low-temp-drone-battery-hacks-stop-sudden-shutdowns-boost-cold-weather-flight-time/> (*IR increase from ~25 mΩ at room temperature to >80 mΩ at -20°C; ~30% capacity attenuation at -20°C*)

HereWin — "Drone LiPo Battery Troubleshooting"

<https://www.herewinpower.com/blog/drone-lipo-battery-troubleshooting-a-comprehensive-guide-to-maintenance-and-safety/> (*New battery IR baseline of 15–20 mΩ; aging threshold of >30 mΩ*)

HobbyKing — "LiPo Battery Low Temperature Performance Analysis and Winter UAV Battery Maintenance Strategies"

https://hobbyking.com/en_us/blog/LiPoBatteryLowTemperaturePerformanceAnalysis (*Electrochemical mechanism of cold-weather IR increase; electrolyte viscosity and ion activity*)

FPV Freedom Coalition — "Beginners Guide to LiPo Batteries"

<https://fpvfc.org/beginners-guide-to-lipo-batteries> (*Practical effects of cold on voltage sag and flight time; over-discharge damage threshold*)

China Hobby Line (CNHL) — "LiPo Internal Resistance Explained"

<https://chinahobbyline.com/blogs/news/lipo-internal-resistance-explained> (*Fresh high-performance cell IR of 1–4 mΩ per cell; state-of-charge and temperature as measurement confounds*)

RCHelicopterFun — "LiPo Internal Resistance — Is Your Battery Still Healthy?"

<https://www.rchelicopterfun.com/lipo-internal-resistance.html> (*Capacity-dependent IR variability; measurement best practices at storage voltage ~3.85V/cell*)

Idea for Motor Diagnostics:

Audio-spectrum motor health check (microphone-based)

- Use a directional mic (or contact mic) placed near each motor while spooling.
- Perform FFT to detect abnormal harmonics/roughness signatures vs a baseline.
- Output: PASS/FAIL + “bearing/imbalance suspected.”
- Why it fits: minimal mechanical coupling, fast scanning on set.

1. Core Idea

Replace the pilot’s subjective “does it sound smooth?” check with:

A fast spectral analysis of motor sound during a controlled spin-up, compared against a known healthy baseline.

Instead of ears detecting irregularities, you use FFT-based frequency analysis.

2. What It Would Actually Measure

When a brushless motor spins, it produces:

- Fundamental frequency (based on RPM)
- Harmonics (multiples of fundamental)
- Broadband noise
- Bearing noise (higher-frequency components)
- Imbalance-related sidebands

Motor faults introduce measurable acoustic changes:

Fault Type	Acoustic Signature
Bearing wear	High-frequency spikes / broadband roughness
Debris in motor	Irregular harmonic distortion
Shaft misalignment	Sidebands around fundamental
Imbalance	Elevated amplitude at fundamental
Winding issues	Abnormal tonal components

Human ears detect obvious roughness.
FFT detects subtle frequency-domain deviations.

QUESTIONS

PHYSICAL DRONE DIMENSIONS _ RESCOPE SIZE REQUIREMENT
SOUND PROOF STUFF

3. How It Would Work (Operationally)

Step 1 — Attach / Position

- Clip-on directional microphone
OR
- Contact mic touching carbon arm

Step 2 — Controlled Spin-Up

- Use flight controller test mode.
- Spin motor from low to moderate RPM (3–5 seconds).

Step 3 — Signal Processing

Device performs:

- Short-time Fourier Transform (STFT) or FFT
- Extract:
 - Peak frequencies
 - RMS amplitude
 - High-frequency energy band
 - Harmonic distortion metric

Step 4 — Compare to Baseline

Two possible methods:

A. Absolute threshold method

- If high-frequency energy > threshold → FAIL
- If harmonic distortion > X% → WARN

B. Baseline comparison method

- Store “healthy” spectral profile for that drone.
- Compute deviation score.
- If deviation > threshold → FAIL

4. Hardware Architecture

Minimal Viable Hardware

- High-sensitivity MEMS microphone (20 Hz–20 kHz minimum)
- Microcontroller (ESP32, STM32, etc.)
- ADC (≥ 44 kHz sampling)
- Small display or LED indicator
- Button + power source

Optional upgrade:

- Contact piezo mic for better signal isolation.

5. What the User Experience Would Look Like

Target: ≤ 2 interactions

Example:

1. Clip mic to motor arm.
2. Press “Test.”

Device automatically:

- Spools motor
- Captures audio
- Processes FFT
- Outputs:

Motor 3: PASS

or

Motor 3: FAIL – Bearing Noise Detected

Time required: ~5–8 seconds per motor.

Most dangerous motor failures in small brushless motors are **bearing-related**.
Bearing defects produce predictable frequency components based on geometry.

The key fault frequencies are:

- BPFO — Ball Pass Frequency Outer race
- BPFI — Ball Pass Frequency Inner race
- BSF — Ball Spin Frequency
- FTF — Fundamental Train Frequency (cage frequency)

These are computed from:

- Shaft speed (RPM)
- Number of rolling elements
- Ball diameter
- Pitch diameter
- Contact angle

SOURCES CHAT GAVE ME _ NOT VERIFIED YET

- Randall, R. B. *Vibration-based Condition Monitoring: Industrial, Aerospace and Automotive Applications*. Wiley, 2011.
This is one of the core references for vibration diagnostics.
- Mobley, R. K. *An Introduction to Predictive Maintenance*. Elsevier.
Very readable overview of fault frequency identification.

Goals needed. Mentioned in meeting with anastasia

Acquire information

Process Information

Relay Information

VIBRATION TEST

Here's a full practical breakdown using the paper's method as the scientific backbone but adapted to the FPV context:

Step 1 — The Vibration Sensor

What the paper used: MFC 8528P2 piezo patch **What you use:** MPU-6050 MEMS accelerometer (or ADXL345 for higher bandwidth)

The MPU-6050 measures acceleration in all three axes digitally over I2C. It's not piezoelectric — it doesn't generate its own voltage — but it measures the same thing: arm vibration amplitude and frequency. The tradeoff is that the MPU-6050 has a max sampling rate of ~1 kHz (just barely enough, per the paper's 1048 Hz) and limited bandwidth (~260 Hz), so it may miss some high-frequency harmonics the paper's piezo patch caught above 300 Hz.

Cost: ~\$2–5 CAD on AliExpress, ~\$10–15 at a local electronics supplier **Where to mount it:** Directly on the motor mount or the arm near the motor — rigid contact with the frame, no air gap. Use double-sided foam tape or a small screw mount.

Step 2 — The Microcontroller

What the paper used: Digilent Analog Discovery 2 + laptop **What you use:** ESP32 (already in your project) or an **Arduino Nano**

The microcontroller reads the MPU-6050 over I2C at the highest possible sample rate, collects 1–2 seconds of data, runs a basic FFT using the [arduinoFFT](#) library, and compares the resulting frequency spectrum against a stored healthy baseline. The ESP32 is preferred over the Nano here because it has significantly more RAM (needed to hold FFT buffers) and runs at 240 MHz (fast enough for real-time FFT).

Cost: ESP32 ~\$5–10 CAD

Step 3 — The Test Protocol (How to Actually Run It)

This is directly adapted from the paper's experimental design:

During the flight of the UAV, damage is initiated with a specific level of intensity. The percent of the PWM denotes the time by which the motor's phase is shut off, leading to interrupted operation. Throughout each case, the duration of the experiment was consistently set at 10 seconds. [PubMed Central](#)

For Danny's pre-flight check, you don't need 10 seconds. The adapted protocol would be:

1. **Drone is stationary** on a flat surface (not hovering — the paper actually confirms ground testing works, as their test was performed while the drone was held in a test stand)
2. **Spin the motor under test to a fixed throttle** — e.g., 30% throttle via a bench ESC tester or through Betaflight's motor test mode
3. **Collect 2–3 seconds of accelerometer data** from the arm at ~500–1000 Hz
4. **Run FFT** on the microcontroller — identify dominant frequency peaks and compare their amplitude and frequency against the baseline recorded from a known-healthy motor
5. **Output pass/fail** — if dominant peak amplitude is within X% of baseline and no unexpected harmonics appear (e.g., a 300 Hz spike that wasn't there before), it's a pass

Step 4 — Establishing a Baseline (Critical Step the Paper Emphasizes)

The experimental baseline condition for no damage condition was established. The damage quantifiers proposed in this paper are useful for rapid monitoring of unmanned aerial vehicle operations in connection with this baseline. [PubMed Central](#)

This means before the system is useful, Danny needs to run each motor in a known-healthy state and **save that FFT profile as the reference**. Every future test compares against that stored fingerprint. Without this step, there's nothing to compare against — you can't just flag "unusual" frequencies without knowing what "normal" looks like for that specific motor.

In your implementation, this would be stored in the ESP32's flash memory as a simple array of expected frequency amplitudes.

Full Shopping List

Component	Purpose	Approx. Cost (CAD)
MPU-6050 breakout board	Vibration sensing on arm	\$5–15
ESP32 development board	FFT processing + output	\$8–15

Small breadboard + jumper wires	Prototyping connections	\$5
Double-sided foam tape / epoxy	Mounting sensor to arm	\$3–5
Bench ESC tester or motor tester	Spin motor at fixed throttle for test	\$10–20
OLED display (0.96" I2C)	Show pass/fail output	\$5–10
Total		~\$36–70 CAD

One Key Limitation to Flag in Your RFP Response

The paper used an octocopter with **aluminium arms** — these flex significantly, making piezo patches ideal. Danny's FPV drone uses **carbon fibre arms**, which are stiffer and transmit vibration differently. This means the specific frequency thresholds from the paper won't transfer directly — they'll need to be re-characterized experimentally for the FPV frame geometry. That's not a fatal flaw, but it's important to acknowledge in your design justification: the sensing *principle* is validated by this paper; the specific *calibration values* require a baseline test on Danny's actual hardware.

Full Shopping List with Links

1. MPU-6050 Accelerometer / Gyroscope (GY-521 Breakout Board)

The core vibration sensor. Buy a multi-pack since they're cheap — you'll want spares for testing and one per motor in a full build.

- **Amazon.ca — 8-pack ~\$13 CAD:**
<https://www.amazon.ca/D-FLIFE-MPU-6050-Accelerometer-Gyroscope-Converter/dp/B08F51SQ71>
- **DigiKey.ca — single chip (bare IC, needs soldering):**
<https://www.digikey.ca/en/products/detail/tdk-invensense/MPU-6050/4038009>

Get the Amazon.ca 8-pack. The GY-521 breakout board already has the chip mounted, pins soldered, and voltage regulation built in — plug and play with an ESP32.

2. ESP32 Development Board

The brain. Runs FFT, stores the baseline profile, outputs pass/fail.

- **Amazon.ca — ELEGOO 3-pack ~\$25 CAD (USB-C, good quality):**
<https://www.amazon.ca/esp32-dev-board/s?k=esp32+dev+board>
- **Direct ELEGOO listing (Type-C):**
<https://www.amazon.ca/Development-Dual-Mode-Microcontroller-ESP-WROOM-32-Integrated/dp/B08HJYT6YV>

The ELEGOO boards are well-regarded in the hobbyist community, come pre-soldered, and work directly in Arduino IDE.

3. 0.96" OLED Display (SSD1306, I2C)

For displaying PASS / FAIL output and FFT peak readings without needing a laptop.

- **Amazon.ca — UCTRONICS single module ~\$10 CAD:**
<https://www.amazon.ca/UCTRONICS-SSD1306-Self-Luminous-Display-Raspberry/dp/B072Q2X2LL>
- **Amazon.ca — Unifun 2-pack ~\$13 CAD (white, cleaner display):**
<https://www.amazon.ca/Display-Module-SSD1306-Compatible-Arduino/dp/B07QW95L1B>

The SSD1306 is the standard for Arduino/ESP32 projects — tons of tutorials and a drop-in library ([Adafruit_SSD1306](#)).

4. Breadboard + Jumper Wires

For prototyping the circuit before any permanent assembly.

- **Amazon.ca — WayinTop full kit (breadboard + assorted wires) ~\$15 CAD:**
<https://www.amazon.ca/WayinTop-Breadboard-Arduino-Preformed-Solderless/dp/B07PTYBFDI>
- **Canada Robotix (local Canadian supplier, 65-wire set):**
<https://www.canadarobotix.com/products/922>

The WayinTop kit is the best value — it includes both rigid U-shaped wires (for clean breadboard work) and flexible Dupont cables (for connecting the sensor to the ESP32).

5. ESC / Servo Tester (to spin motor at fixed throttle during test)

This lets you spin a single FPV motor at a controlled, repeatable throttle level without needing a full flight controller or radio transmitter. You plug it between a LiPo battery and the ESC.

- **Amazon.ca — 3CH ESC/Servo Tester with built-in BEC ~\$12 CAD:**
<https://www.amazon.ca/Centering-Tester-Precision-Airplanes-Control/dp/B08H223QT9>
- **Amazon.ca — Multi Servo/ESC Tester ~\$10 CAD:**
<https://www.amazon.ca/Tester-Electronic-Controller-Consistency-Checker/dp/B0788RPBV5>

Either works. You plug your ESC into the tester, turn the knob to ~30% throttle, and the motor spins at a consistent, repeatable speed — exactly the controlled condition the paper used.

6. Double-Sided Foam Tape (for mounting sensor to arm)

Not glamorous, but critical. The sensor needs *rigid* contact with the arm to transmit vibrations accurately. Foam tape provides damping — you actually want a thin, firm layer, not a thick squishy one.

- **Canadian Tire or Home Depot locally** — 3M brand double-sided mounting tape (thin, firm variety), ~\$5–8 CAD. Alternatively, a small drop of epoxy or hot glue on the sensor housing works even better for a permanent mount.
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Total Estimated Cost

Item	Source	Est. Cost (CAD)
MPU-6050 8-pack	Amazon.ca	~\$13

ESP32 3-pack	Amazon.ca	~\$25
OLED Display	Amazon.ca	~\$10
Breadboard + Wires kit	Amazon.ca	~\$15
ESC/Servo Tester	Amazon.ca	~\$12
Mounting tape/epoxy	Canadian Tire	~\$6
Total		~\$81 CAD

This is well within a first-year prototype budget, and most of these components (the ESP32, breadboard, wires, OLED) are reusable across the battery and RF modules you've already designed. The marginal cost just for the vibration module specifically is really only the MPU-6050 (~\$2 each) and the ESC tester (~\$12).

N

Notes

Check equipment before shoot

2-3min window to reset the drone

2-3min for diagnostics

Memory on the board already that give blackbox data (USB C)

Analyze black box data?

30-60min down time to check black box data

Motor output, gyro output, loose wire on camera?

Sound doable

Replace when change different frames

Maintenance not predictable, never know

Motor, boards, battery that they buy

Charging and checking battery (wanna see resistance in battery, higher resistance -> worse battery)

yes/no, look into the problem afterwards

Accelerometer, gyroscope

Parameter ?

Magnetometer? no

GPS ? no

Bareboned

To know it's working or not,

Vibration reading given by black box

Isolate motor: patches

Use black box to tune the drone

Pin loops, PIDs

Implementation Process

- | | |
|--|---|
| 1. Gather the hardware | You need: an ACS712 20A Hall-effect current sensor module — specifically the 20A variant, not the 5A version, because the GEPRC SPEEDX2 1404 peaks at 14.3A. Also: an ESP32 development board, two resistors (1 k Ω and 2 k Ω) for a voltage divider, and a USB cable. No modification to the motor circuit is required, though wiring the sensor module into the signal chain does involve making electrical connections. |
| 2. Thread one motor phase wire through the sensor | A BLDC motor has three thin wires running from the ESC to the motor bell — the phase wires (A, B, C). Thread one of them through the hole in the ACS712 module. The sensor does not cut or modify the wire; it senses the magnetic field created by current flowing through it. Connect the sensor's signal output to the ESP32's GPIO34 pin through the voltage divider (steps down 5 V to 3.3 V for the ESP32). |
| 3. Spin the motor at controlled throttle via Betaflight | Remove all propellers. Connect the drone via USB to a laptop running Betaflight Configurator. Navigate to the Motor Test tab. Set one motor to 20% throttle — for the SPEEDX2 1404 4600KV on 4S (~16.8V), this produces approximately 15,500 RPM and a commutation frequency of ~1,550 Hz, well within the sensor's detectable range. Keep the other three motors at zero. Hold for 3–5 seconds at steady speed. |

4. Sample the current signal at high speed on the ESP32

While the motor spins, the ESP32 samples the ACS712 output at approximately 50,000 samples per second (50 kHz). ESC switching frequencies typically fall in the 20–48 kHz range; sampling below this would alias switching artefacts directly into the fault frequency band. A 50 kHz sample rate (Nyquist limit: 25 kHz) provides margin above this range while remaining within the ESP32's ADC capability. The ESP32 stores the time-series in memory and transmits it to the laptop via USB serial.

5. Run FFT on the laptop

A short Python script (using NumPy) receives the current data and applies a Fast Fourier Transform (FFT). This converts the signal from time-domain (current vs time) to frequency-domain (amplitude vs frequency). A healthy motor shows only the expected electrical frequency harmonics. Faults appear as sideband peaks at predictable positions around each harmonic, described by: $f_{\text{fault}} = f_e \pm k \times f_m$, where f_e is the electrical frequency (~1,550 Hz at test throttle), f_m is the mechanical rotation frequency (~258 Hz), and k is the harmonic index (1, 2, 3...). Each fault type shifts energy to characteristic sideband positions in this spectrum.

6. Compare against the stored healthy baseline

The first run on a confirmed-healthy motor saves the spectrum as a baseline. On each subsequent pre-flight check, the new spectrum is compared to the normalised baseline. Peaks exceeding a threshold — set empirically during baseline capture, accounting for measurement noise — trigger a fault flag. Determining appropriate thresholds and noise floor characterisation is an expected validation step for a prototype. Repeat for all four motors in sequence — estimated total bench time: 2–3 minutes.

Questions to Ask

1. Full hardware stack of the community's drone currently. Type of radio transmitters and receivers used, battery used, video receiving, drone telemetry sensors (if any) etc.
2. Physical Drone Dimensions
3. Solution size requirement rescoping? Ask the community if it can be any bigger?
4. How strict is the 30 second time constraint
5. Would you rather be given a yes or no decision on if the drone is safe to fly or more in depth data for the drone operator to analyze
6. During the last interview you had you mentioned not more than two
7. Drone replacement cycle

Voltage Measurement Prototype

1. Figure out what kind of threshold are we looking for in the data to know if this will show problems with the motor
2. Get a multimeter with usb interface to read the data from motor of the drone
3. Code will read data in and pre process into a useable format
4. Code will display an output graph of what the data looks like (Voltage vs. Time graph)
5. Output all the pre processed data for debugging and error checking purposes

Sound Measurement

1. Record audio file through phone and import to python
2. Use python librosa or pydub libraries to read the audio file
3. Make an output graph of the varying frequencies
4. Make a sound damping box for recording the audio from the drone

Sound Proof Box

